

theft, cloning and over-building, to name a few."

Let us take a look at the advancements and additions, which are not only making today's FPGA development boards cost-effective and powerful, but also flexible to work with.

Advancements in FPGA based systems

FPGA platforms are currently being used for high-performance cloud computing, network processing, signal processing, low-latency and data acquisition applications. Dhiraj Kumar, director, Systems Engineering, Argus Technologies, says, "The adaptation of advanced process technologies by FPGA vendors has resulted in the launch of Virtex UltraScale, Arria 10 and Stratix 10 products that support a multi-node portfolio to address a wide set of applications on FPGA platforms."

He adds, "Further advances in FPGA architecture and system in package (SiP) technology are able to address device-level input and output (I/O) bottlenecks and optimisation of sequential applications to offer platform-level integration of heterogeneous technologies with remarkable speed-ups."

FPGA providers are keen on achieving higher road maps in a shorter time period. At a utilisation rate of 60 per cent, single FPGA devices can hold up to 20 million application-specific integrated circuit (ASIC) gates, informs Sudharsan Palanikumar, senior analyst, Femto Logic Design Pvt Ltd. He says, "The key advantage of an FPGA based system is that it can achieve higher speeds without any hassles."

He adds, "The new era of FPGA systems can hold up to 100 million gates that help bring down the size of a larger application considerably."

Today's FPGA boards are also increasingly focused on high-speed communication infrastructure such as multi-gigabit serial transceivers, integrated computing and digital signal processing (DSP) resources to cater

SoCs, FPGAs and the need for SoC based FPGAs

There is a great saying, "Advancement in recent trends would not halt for any man." Difficulties in fabricating integrated circuits (ICs) for FPGA based prototyping boards have become more complex due to size, in which porting 100 million ASIC gates takes months and also needs several programmable devices. Keeping all hurdles apart, FPGA providers are keen on delivering higher-grade FPGAs in terms of capacity and I/Os.

In a parallel track, SoC projects are growing much faster. Adding an extra grade of complexity to the system, processor sub-systems in SoC designs are developed using virtual platforms, which gives developers/customers the option to switch between various processor configurations based on the end user requirement. From a designer's perspective, complexities in developing these systems are tricky. When such systems are sent for verification, it takes enormous time and engineering resources (hardware and software tools). In certain cases, modular-verification methods fail abruptly for larger design files and sophisticated design constraints.

As we see it, the need for developing independent sub-systems for the Internet of Things (IoT) architecture demands pure SoC based FPGAs for fixed general-purpose control sub-systems and data-acquisition sub-systems. There is always a recognisable demand for FPGA based prototypes systems, but the time-to-market makes these a secondary choice by developers. In complex sub-systems, the design demands careful portioning and timing optimisation steps, where FPGA based prototypes take a much longer time to become available until silicon is back.

The debugging feasibility into the hardware is quite a tough task when compared with simulation and software based emulation. In such cases, hardware probes can be inserted, which, in turn, reduces the speed grade of the system. Hence, FPGA based prototypes have come to the limelight by adoption at the final stages of development, during which register-transfer level (RTL) has already been verified and deemed stable.

—Contributed by Sudharsan Palanikumar, senior analyst, Femto Logic Design Pvt Ltd

to the growing communication, medical and aerospace markets, informs Anoop Jose, manager - research and development, Numato Systems Pvt Ltd. He says, "This change is very visible in the new FPGA offerings from various leading companies. Some top-of-the-series devices offer serial bandwidth in excess of 5Tb/s."

He adds, "A large number of FPGAs offer integrated PowerPC/ARM hard processor units and a large array of DSP slices to cater for the ever-growing demand for signal processing and communication." Although, a fair amount of development boards are slowly showing up in the market. These facilitate easy FPGA based solution development using the above-mentioned technologies.

Current FPGA platforms are able to offer much better performance than some of the fastest graphics cards available today, addressing central processing unit (CPU)/graphics processing unit (GPU) markets as well, informs Kumar. He says, "High-end FPGAs are playing a dominating role in military and aerospace

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communications infrastructure and advanced radar applications."

Additions in FPGA developments boards

With over 15 years of experience of creating FPGA based development boards, Avnet had introduced their Mini-Module series designed as a systems on module (SoM) more than ten years ago. It allowed an engineer to simply use the development board as part of the final product.

A few years ago, Xilinx introduced Zynq-7000, an all-programmable system on chip (SoC), which was a hybrid chip having both a processing system as well as FPGA program-